

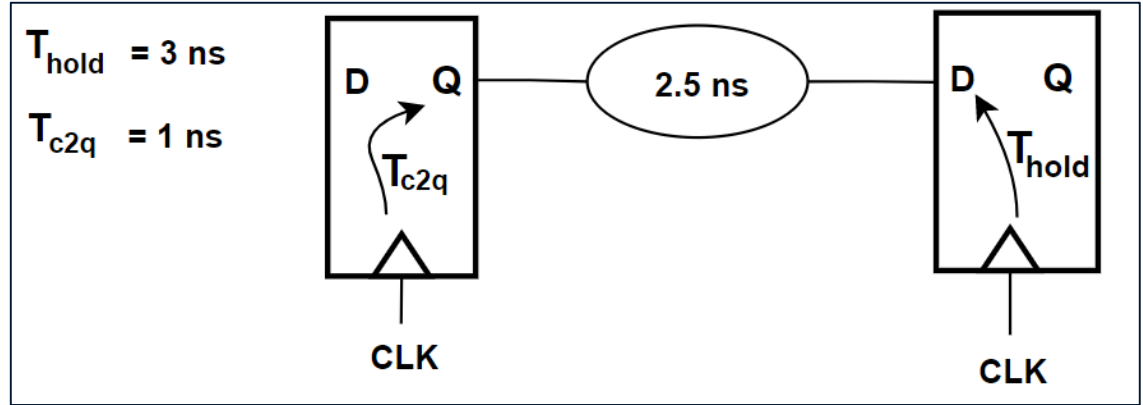
Example 1

- **Assumptions**

1. Interconnect Delays not considered
2. Clock Uncertainty not considered
3. One Operating Condition

- **Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack



Solution

Hold Analysis

$$\text{Data Arrival Time} = T_{cq} + T_{pd} = 1 + (2.5) = 3.5$$

$$\text{Data Arrival Time} > \text{Hold Time} .$$

No Violation

$$\text{Slack} = \text{Data Arrival Time} - \text{Hold Time} = 3.5 - 3 = 0.5$$

Example 2

- **Assumptions**

1. Interconnect Delays not considered
2. **Clock Uncertainty is considered**
3. One Operating Condition

- **Required**

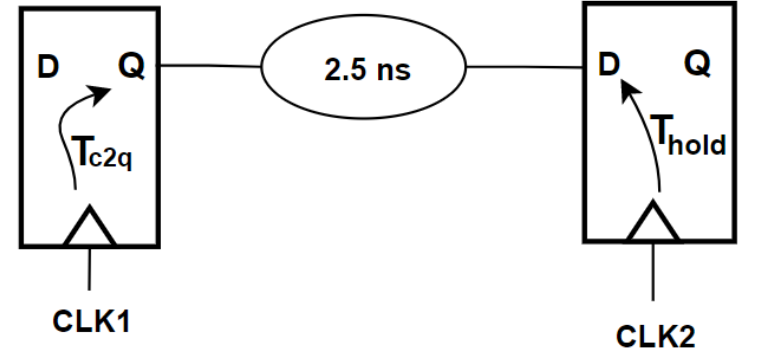
1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack

$$T_{\text{hold}} = 3 \text{ ns}$$

$$T_{\text{c2q}} = 1 \text{ ns}$$

Skew

$$\text{CLK 1_2} = 0.7 \text{ ns}$$



Solution

Hold Analysis

$$\text{Data Arrival Time} = T_{cq} + T_{pd} = 1 + (2.5) = 3.5$$

For Postive Skew

$$\text{New Hold Time} = \text{Skew} + \text{Hold Time} = 0.7 + 3 = 3.7$$

$$\text{Data Arrival Time} < \text{New Hold Time}$$

Violation

$$\text{Slack} = \text{Data Arrival Time} - \text{New Hold Time} = 3.5 - 3.7 = -0.2$$

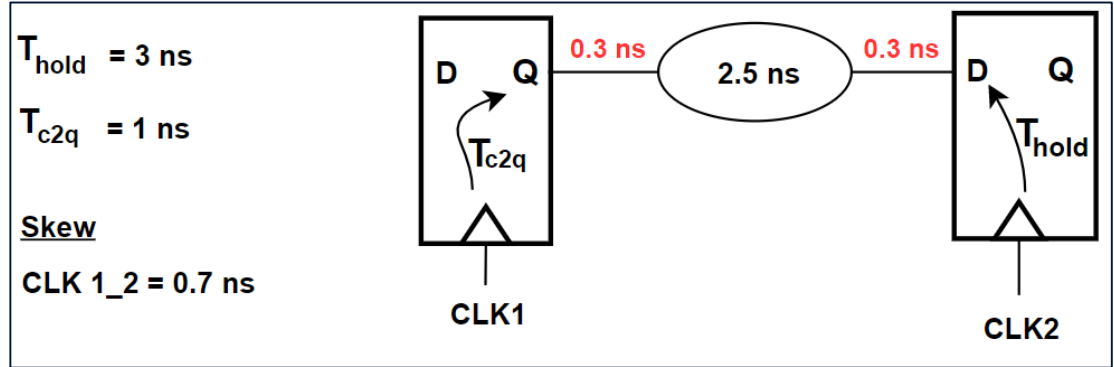
Example 3

- **Assumptions**

1. Interconnect Delays is considered
2. Clock Uncertainty is considered
3. One Operating Condition

- **Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack



Solution

Hold Analysis

$$\text{Data Arrival Time} = T_{cq} + T_{pd} = 1 + (0.3 + 2.5 + 0.3) = 4.1$$

For Postive Skew

$$\text{New Hold Time} = \text{Skew} + \text{Hold Time} = 0.7 + 3 = 3.7$$

$$\text{Data Arrival Time} > \text{New Hold Time}$$

No Violation

$$\text{Slack} = \text{Data Arrival Time} - \text{New Hold Time} = 4.1 - 3.7 = 0.4$$

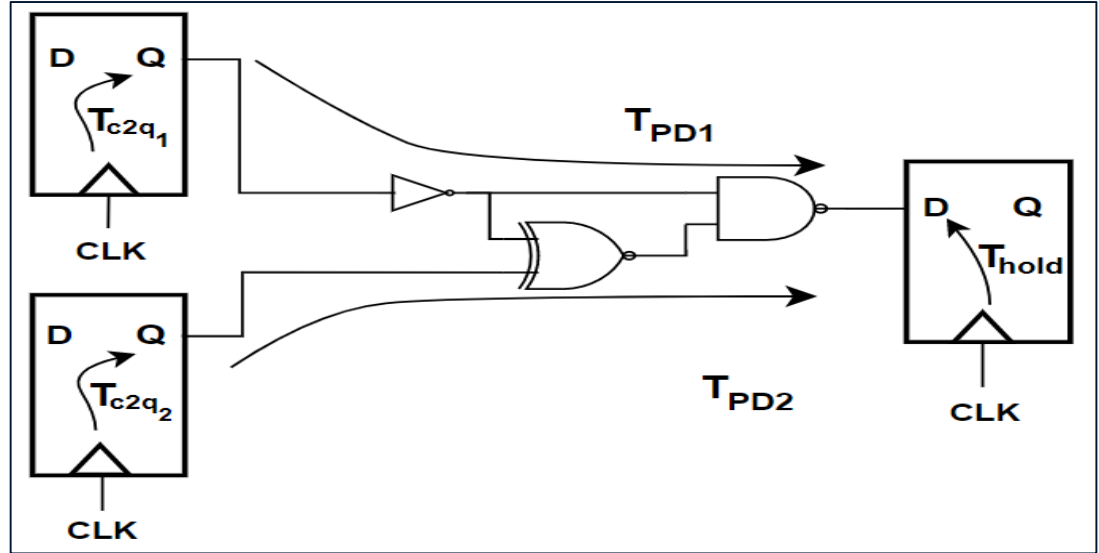
Example 4

- Assumptions**

1. Interconnect Delays is not considered
2. Clock Uncertainty is not considered
3. **Worst/Best Operating Conditions**

- Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack
3. if not calculate the negative slack and suggest a solution to fix hold violation



Delays	Inverter	XNOR	NAND	Tcq1	Tcq2	Thold
Max	0.5	1.5	1	0.4	0.35	2.2
Min	0.45	1.3	0.95	0.3	0.3	2

Solution

Hold Analysis

PATH 1

$$\text{Data Arrival Time} = T_{cq1(min)} + T_{pd1(min)} = 0.3 + (0.45 + 0.95) = 1.7$$

$$\text{Data Arrival Time} < \text{Hold Time}$$

Violation

$$\text{Slack} = \text{Data Arrival Time} - \text{Hold Time} = 1.7 - 2 = -0.3$$

Solution

Hold Analysis

PATH 2

$$\text{Data Arrival Time} = T_{cq1(min)} + T_{pd11(min)} = 0.3 + (0.45 + 1.3 + 0.95) = 3$$

$$\text{Data Arrival Time} > \text{Hold Time}$$

No Violation

$$\text{Slack} = \text{Data Arrival Time} - \text{Hold Time} = 3 - 2 = 1$$

Solution

Hold Analysis

PATH 3

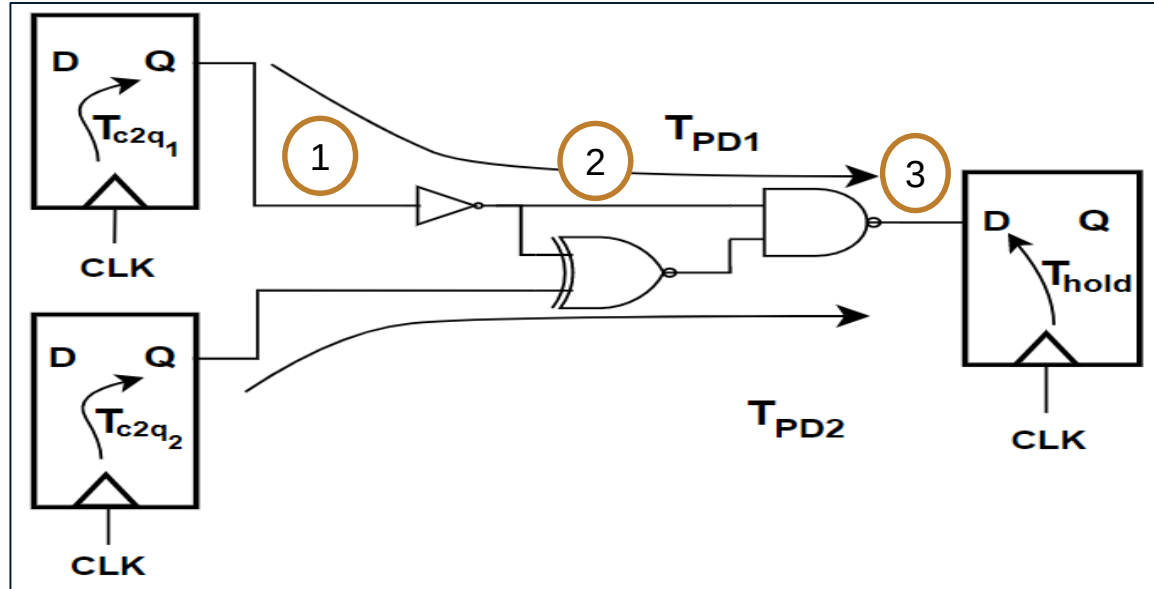
$$\text{Data Arrival Time} = T_{cq2(min)} + T_{pd2(min)} = 0.3 + (1.3 + 0.95) = 2.55$$

$$\text{Data Arrival Time} > \text{Hold Time}$$

No Violation

$$\text{Slack} = \text{Data Arrival Time} - \text{Hold Time} = 2.55 - 2 = 0.55$$

Example 4



Fix : Add Buffer at point 2 between inverter and NAND